

TrustLayer: Judge Q&A; Guide

VIKSIT BHARAT IDEATHON 2026 • CYBER SAFETY
First-year CSE team preparation

Core positioning: TrustLayer is an evidence-based digital media verification platform. It examines AI-detection signals, provenance, metadata and forensic indicators, then presents the findings as a transparent assessment.

Golden rule: *Never claim 100% detection or guaranteed authenticity. Say **assessment, evidence, signals, confidence**, and **requires review**. Clearly distinguish what is implemented in the prototype from future scope.*

1. Product & Problem

Q. What exactly is TrustLayer?

Answer: TrustLayer is a multi-signal digital media verification platform. A user submits an image or video, and the system examines several evidence layers, including AI-generation indicators, provenance, metadata and forensic signals, then presents the findings in a structured report.

Q. What problem are you solving?

Answer: People increasingly encounter AI-generated and manipulated media but usually lack a simple way to inspect its origin or technical evidence. TrustLayer addresses this verification gap in one workflow.

Q. Why is this a Cyber Safety problem?

Answer: Manipulated media can support impersonation, scams, social engineering, misinformation and reputational harm. TrustLayer helps users inspect suspicious media before trusting or sharing it.

Q. Why not just call it a deepfake detector?

Answer: Detection is only one source of evidence. Provenance, metadata and forensics answer different questions, so TrustLayer combines multiple signals rather than relying on one opaque detector.

Q. Who would use it?

Answer: Potential users include everyday users, journalists and fact-checkers, researchers and security teams. The hackathon prototype focuses on a simple verification workflow.

2. Verification Method

Q. What happens after upload?

Answer: The intended flow is Upload → Analyze → Review Evidence → Understand Risk. The system runs multiple checks and presents the findings in a structured report.

Q. What are the four evidence layers?

Answer: AI Detection looks for synthetic-content indicators. Provenance checks available content credentials or provenance information. Metadata reviews embedded file information. Forensics examines visual or temporal inconsistencies.

Q. What is provenance?

Answer: Provenance describes where content came from and how it was created or edited. C2PA provides a technical standard for supported content provenance and Content Credentials.

Q. Does missing provenance mean fake?

Answer: No. It only means that this evidence source is unavailable or unverifiable. That is why TrustLayer uses multiple signals.

Q. Can metadata prove authenticity?

Answer: No. Metadata can be missing, changed or stripped. It is one evidence layer, not proof.

Q. How would video verification work?

Answer: A video can be sampled into representative frames, analyzed individually and compared across frames. More advanced temporal analysis can be added as the system evolves.

3. AI / ML Questions

Q. What AI model are you using?

Answer: The proposed architecture uses Python and PyTorch for ML components, with OpenCV and FFmpeg for media processing. The detector is modular so research models can be replaced or upgraded.

Q. How do you know media is AI-generated?

Answer: We do not rely on one clue. A detector can look for learned synthetic patterns while provenance, metadata and forensic checks provide additional evidence. The output is an assessment, not certainty.

Q. Can it detect every new generator?

Answer: No. Generalization to unseen generation methods is a known challenge. TrustLayer is designed so new models and evidence sources can be added over time.

Q. What if the detector is wrong?

Answer: The detector is only one signal. Other layers provide context, and conflicting or weak evidence should lead to a review-oriented assessment rather than an absolute claim.

Q. Why multiple signals?

Answer: Each signal answers a different question: synthetic patterns, content history, file information and visual or temporal consistency. Combining them gives a more transparent evidence picture.

4. Technology & Architecture

Q. What is your tech stack?

Answer: Frontend: Next.js/React and Tailwind CSS. Backend: Python with FastAPI. AI/ML: PyTorch. Media processing: OpenCV and FFmpeg. Provenance: C2PA. Storage can use Supabase/PostgreSQL. These are proposed implementation technologies for the prototype.

Q. Why Next.js?

Answer: It provides a strong responsive web foundation, clean application structure and straightforward deployment to platforms such as Netlify.

Q. Why FastAPI?

Answer: It is lightweight and well suited to Python AI and computer-vision services. It creates a clean API boundary between the interface and analysis engine.

Q. Why Python?

Answer: Python has a strong ecosystem for machine learning and computer vision, making it practical for integrating research models and media-processing tools.

Q. Why C2PA?

Answer: C2PA provides a standards-based way to inspect content provenance and Content Credentials when supported by the media.

Q. How can it scale?

Answer: Keep each analysis layer modular. Start with images and a small set of checks, then add video, stronger detectors, browser integration and APIs without redesigning the whole product.

5. Security & Privacy

Q. How would you protect uploaded files?

Answer: For production, use encrypted transport and storage, strict retention rules, access controls and isolated processing. Only claim controls that are actually implemented in the prototype.

Q. Could an attacker upload a malicious file?

Answer: Yes, uploads are an attack surface. Production hardening should include file-size limits, MIME validation, safe parsers, sandboxed processing, resource limits, rate limiting and least-privilege execution.

Q. What if signals disagree?

Answer: Show the disagreement instead of hiding it. The report can identify conflicting evidence and mark the result as requiring review.

Q. Can TrustLayer guarantee authenticity?

Answer: No. A file alone cannot always establish authenticity. TrustLayer provides evidence and an assessment to support safer decisions.

6. Feasibility, Impact & Future

Q. Can first-year students build this?

Answer: Yes, with a focused MVP. We do not need to solve every deepfake problem. A realistic prototype can demonstrate image upload, defined verification checks, evidence aggregation and a clear report.

Q. What is the MVP?

Answer: Upload an image → run selected checks → display evidence → provide an understandable assessment. Video and additional research models remain modular extensions.

Q. What is innovative?

Answer: The key innovation is the product-level combination of different evidence sources into one transparent verification workflow. We are not claiming to invent a new deepfake model.

Q. How will you measure performance?

Answer: Evaluate detectors on labeled datasets using precision, recall and false-positive behavior, and test generalization on unseen generation methods. Also evaluate whether users understand the report.

Q. What comes next?

Answer: Stronger models, robust video analysis, browser extension, multilingual support, API integration, broader provenance support and evaluation across diverse real-world datasets.

7. Research & Credibility

Q. What research supports the idea?

Answer: The presentation references synthetic-content detection and generalization research, including FaceForensics++ and DeepfakeBench, plus research showing challenges with unseen synthetic media.

Q. What is FaceForensics++?

Answer: A benchmark and dataset used to study detection of manipulated facial imagery and evaluate manipulation-detection methods.

Q. What is DeepfakeBench?

Answer: A benchmark framework for deepfake detection research that helps standardize datasets and evaluation of detection methods.

Q. Why NIST?

Answer: NIST's synthetic-content work discusses approaches such as provenance, watermarking, detection, labeling and transparency, supporting the broader multi-signal framing.

Q. Why mention existing systems such as TrueMedia and Diopter AI?

Answer: They provide useful market and research context. Present them as related efforts, not as proof that TrustLayer already has every capability they offer.

8. Tough Judge Questions

Q. Isn't this already available?

Answer: Individual detection and provenance tools already exist. Our focus is a unified, understandable workflow that combines evidence sources and explains why an assessment was produced.

Q. What if social platforms strip metadata?

Answer: That is why metadata is not the only layer. TrustLayer can still use other available signals, such as detection, provenance when present and forensic indicators.

Q. What if someone deliberately tries to fool TrustLayer?

Answer: Adversarial manipulation is a real challenge. The system must be tested against modified and unseen media, and its models and forensic methods should be updateable.

Q. Why should users trust TrustLayer?

Answer: Because it should show evidence instead of only presenting a mysterious score: which signals were available, what was detected, what was missing and how the assessment was formed.

Q. Explain TrustLayer in one sentence.

Answer: TrustLayer helps users assess suspicious images and videos by combining AI detection, provenance, metadata and forensic signals into one transparent evidence-based report.

30-Second Rapid Revision

Problem	AI-generated and manipulated media is increasingly difficult for ordinary users to assess.
Solution	Combine multiple evidence signals into one understandable verification report.
Four layers	AI Detection • Provenance • Metadata • Forensics
Differentiator	Evidence aggregation and transparency, rather than a single opaque detector.
MVP	Upload → Analyze → Review Evidence → Understand Risk
Stack	Next.js/React • FastAPI • PyTorch • OpenCV/FFmpeg • C2PA • Supabase/PostgreSQL
Limitation	No detector is perfect; TrustLayer provides an assessment, not absolute certainty.
Future	Better models • Video • Browser extension • Multilingual support • APIs
One-line pitch	TrustLayer helps users assess suspicious media using multiple independent signals and transparent evidence.

Answering technique: Give the direct answer first. Add one technical detail. If it is future scope, say so explicitly. Never overclaim.

Basis: TrustLayer hackathon materials and architecture discussed in this conversation. Proposed implementation details are labeled as proposed rather than completed.